

Example 15

From the top of a tower 80 m high, two boats are seen in a direction due south. The angles of depression of the boats from the top of the tower are 45° and 30° . Find the distance between the boats.

First, draw a sketch. In Fig. 1.5, $\overline{HT}$ represents the tower; A and B are the positions of the boats.

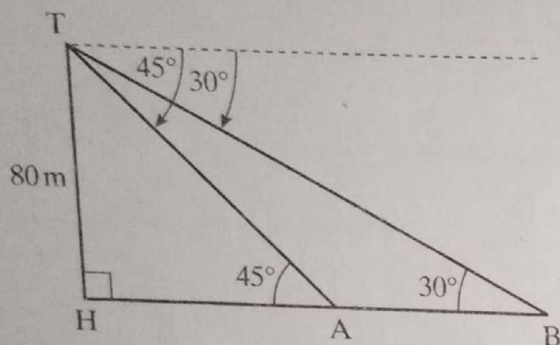


Fig. 1.5

$|AB|$ is the distance between the boats.

$$|AB| = |HB| - |HA|$$

In $\triangle THB$,

$$\tan 30^\circ = \frac{80}{|HB|}$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{80}{|HB|}$$

$$\Rightarrow |HB| = 80\sqrt{3} \text{ m}$$

In $\triangle THA$,

$$\tan 45^\circ = \frac{80}{|HA|}$$

$$\Rightarrow 1 = \frac{80}{|HA|}$$

$$\Rightarrow |HA| = 80 \text{ m}$$

$$\begin{aligned} \Rightarrow |AB| &= 80\sqrt{3} - 80 \text{ m} \\ &= 80(\sqrt{3} - 1) \text{ m} \end{aligned}$$

Notice the importance of drawing a sketch.

Exercise 1k

Draw a sketch in each question. Leave the answers in surd form with rational denominators.

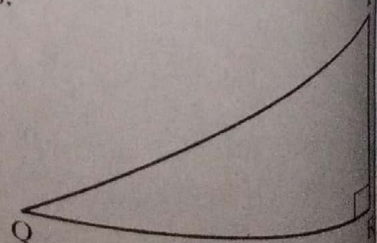
- From the top of a tower, the angle of depression of a boat is 30° . If the tower is 20 m high, how far is the boat from the foot of the tower? [WAEC]

- The angle of elevation of X from Y is 30° . If $|XY| = 40 \text{ m}$, how high is X above the level of Y? [WAEC]
- In $\triangle ABC$, $|AB| = 6 \text{ cm}$, $\hat{A} = 90^\circ$ and $\hat{B} = 30^\circ$, $\overline{AD}$ is an altitude. Find $|CD|$.
- In the isosceles triangle ACB, $|AB| = |AC| = 4 \text{ cm}$, $\hat{BAC} = 30^\circ$ and $\overline{CN}$ is an altitude. Find $|BN|$.
- XYZ is an isosceles triangle with $|XY| = |XZ| = 6 \text{ cm}$ and $\hat{YXZ} = 120^\circ$. Calculate the length of $\overline{YZ}$. [WAEC]
- A cone has a circular base, a perpendicular height of 21 cm, and a semi-vertical angle of 30° . Calculate the slant height of the cone. Find the area of its base. (Take π to be $\frac{22}{7}$.)
- When the angle of elevation of the sun is 30° the shadow of a vertical tower is 20 m longer than when the elevation of the sun is 60° . Find the height of the tower.
- Two huts and a radio mast are on level ground such that one hut is due east of the mast and the other is due west of it. From the top of the mast the angles of depression of the huts are 60° and 45° , respectively. If the mast is 150 m high, find the distance between the huts.
- A and B are two points on level ground, both due south of a flagpole. The angle of elevation of the top of the flagpole is 60° from A and 45° from B. If A is 20 m from the foot of the flagpole, find $|AB|$.
- The top of a building 24 m high is observed from the top and from the bottom of a vertical tree. The angles of elevation are found to be 45° and 60° , respectively. Find the height of the tree.

1.7 Pythagoras' theorem

In a right-angled triangle, the square on the hypotenuse is equal to the sum of the squares on the other two sides.

Fig. 1.6



In Fig. 1.6,

$$PQ^2 = PR^2 + QR^2$$

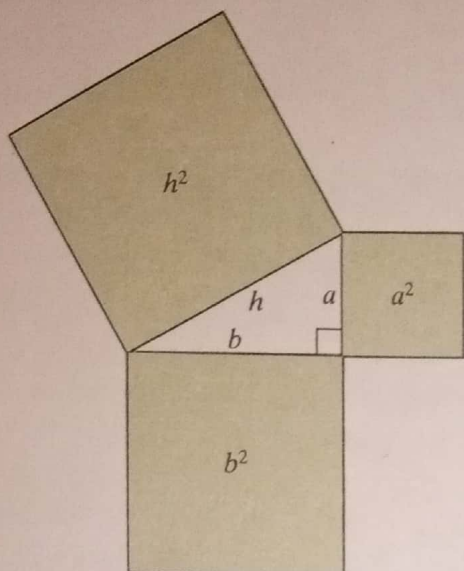


Fig. 1.7

In Fig. 1.7, the triangle has sides of lengths h , a and b units.

$$h^2 = a^2 + b^2$$

also:

$$a^2 = h^2 - b^2$$

$$b^2 = h^2 - a^2$$

Example 16

In Fig. 1.8, $|AC| = 11$ cm, $|BC| = 4$ cm and $|CD| = 10$ cm. Find $|AD|$.

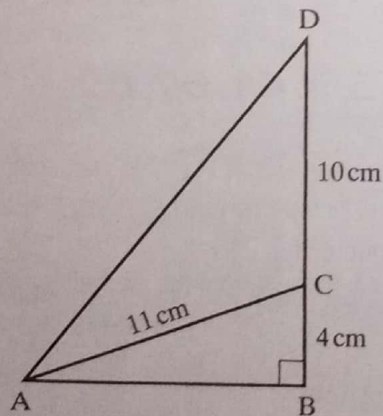


Fig. 1.8

In $\triangle ABC$,

$$\begin{aligned} AB^2 &= AC^2 - BC^2 \\ &= 11^2 - 4^2 \\ &= 121 - 16 = 105 \end{aligned}$$

In $\triangle ABD$,

$$\begin{aligned} AD^2 &= AB^2 + BD^2 \\ &= 105 + 14^2 \\ &= 105 + 196 = 301 \\ |AD| &= \sqrt{301} \text{ cm} \end{aligned}$$

Exercise 11

In this exercise, leave all answers in surd form, simplified as far as possible.

- ① The dimensions in Fig. 1.9 are all centimetres. In each case, x is a whole number. Find the value of x in each part.

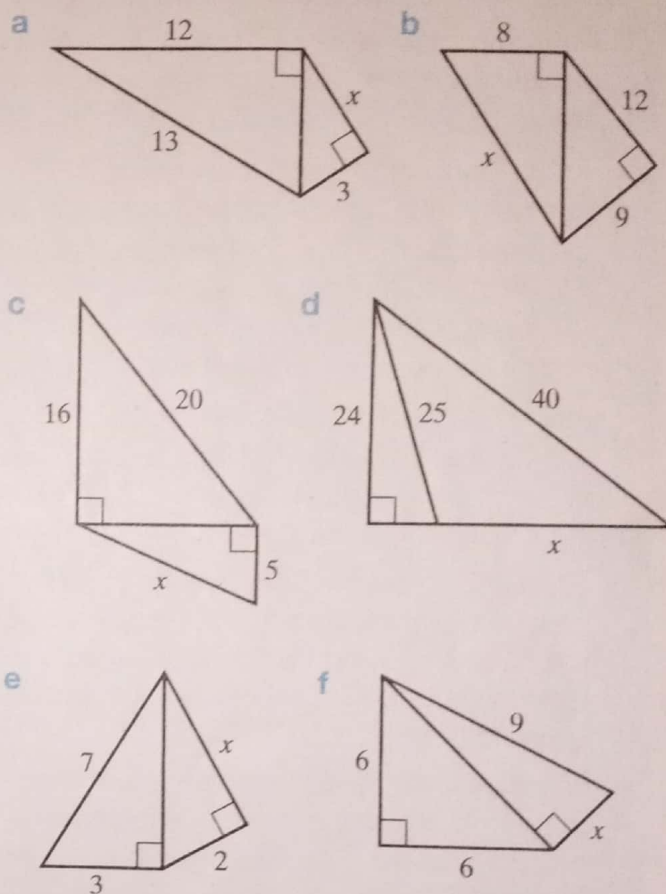


Fig. 1.9

- ② The dimensions in Fig. 1.10 are all centimetres. Find the value of x in each part.

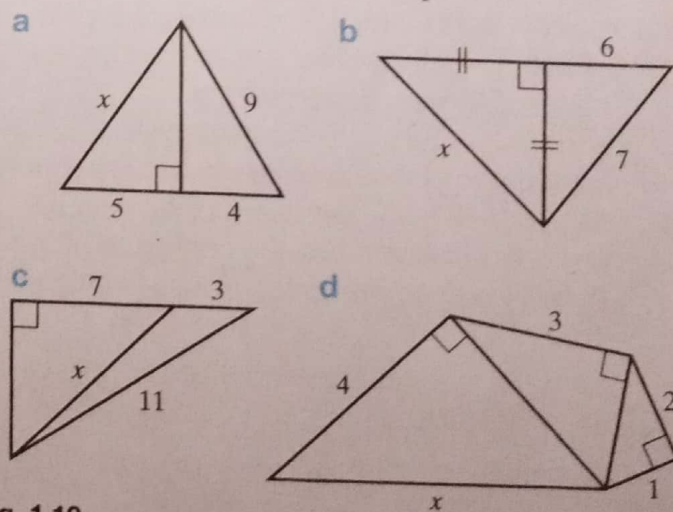


Fig. 1.10

- ③ Find the perpendicular height of a cone which has a slant height of 10 cm and a circular base of diameter 10 cm.

- 4 A vertical radio mast is supported by wires attached to its top and to points on the level ground 9 m from its base. If each wire is 31 m long, find the height of the mast.
- 5 Village B is 7 km due south of village A and village C is 9 km due south of B. Village D is 8 km from A and is due east of B. Find the distance of D from: **a** B **b** C.

SUMMARY

- 1 A *surd* is an irrational or *non-rational* number of the form $a\sqrt{n}$, where n is not a perfect square. Thus, for example, $\sqrt{2}$, $2\sqrt{3}$, $\sqrt{10}$ and $-2\sqrt{5}$ are surds but $\sqrt{4}$, $2\sqrt{9}$, $\sqrt{100}$ are not. *Irrational* numbers cannot be expressed as ratios in the form $\frac{a}{b}$.
- 2 Like surds have the same numbers under the square root sign: $2\sqrt{5}$ and $6\sqrt{5}$ are like surds. We can add and subtract like surds. For example, $10\sqrt{m} - 7\sqrt{m} - \sqrt{m} = 4\sqrt{m}$.
- 3 To *simplify a surd* usually means to find an equivalent surd in which the number under the square root sign is as small as possible. Examples 1 and 3–5 show various forms of simplification.
- 4 To *rationalise the denominator* means to make the denominator of an expression a rational number. See Examples 6–8.
- 5 Clear brackets with surds by following the rules of algebra. See Example 9.
- 6 A *binomial surd* contains two terms, e.g. $(3 + \sqrt{2})$ and $(5\sqrt{3} - 3\sqrt{5})$ are binomial surds. One binomial surd is the *conjugate* of another, if they differ only in the sign connecting the two terms, e.g. $(3 - \sqrt{2})$ and $(5\sqrt{3} + 3\sqrt{5})$ are the respective conjugates of the binomials in the previous sentence. If a fraction has a binomial surd in its denominator, use the conjugate of the denominator to rationalise it. See Examples 10 and 11.
- 7 Some of the trigonometric ratios of special angles can be expressed in surd form.

$$\cos 30^\circ = \frac{\sqrt{3}}{2}$$

$$\tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$\sin 45^\circ = \frac{1}{\sqrt{2}}$$

$$\cos 45^\circ = \frac{1}{\sqrt{2}}$$

$$\sin 60^\circ = \frac{\sqrt{3}}{2}$$

$$\tan 60^\circ = \sqrt{3}$$

Challenge Box

Some sums

	4	3	2
+	6	5	7
	1	0	8
			9

Ten cards, numbered 0 to 9, are arranged to make a correct addition sum.

Find other ways of arranging the ten cards so that two 3-figure numbers add correctly to give a 4-figure number.

Challenge Box

Primes

Here is a set of six prime numbers which use the digits 1 to 9 exactly once.

2, 3, 5, 41, 67, 89

Find more ways of doing this (i.e. make other sets of six prime numbers using the digits 1 to 9 once only)

OBJECTIVES

By the end of this chapter you should be able to:

- 1 Express statements given in index form (such as $81 = 3^4$) as an equivalent logarithm statement ($\log_3 81 = 4$)
- 2 Evaluate expressions given in logarithmic form
- 3 Note the equivalence between the laws of indices and the laws of logarithms
- 4 Recall and use the laws of logarithms to simplify and/or evaluate given expressions without the use of logarithm tables
- 5 Use logarithm tables for the purposes of calculation.

Teaching and learning materials

Teacher: Chart demonstrating laws of logarithms; logarithm tables (as used by Examinations Boards)

Student: Logarithm and antilogarithm tables (see pages 272–273 of this book)

2.1 Logarithms and indices

Books 1 and 2 of *New General Mathematics for Senior Secondary Schools* showed that the statements $100 = 10^2$ and $\log 100 = 2$ are two ways of writing the same thing. The second statement in full is $\log_{10} 100 = 2$, i.e. 'the logarithm to the base 10 of 100 is 2'.

Logarithms can be in bases other than 10. For example, since $8 = 2^3$, $\log_2 8 = 3$ or, in words, the logarithm to the base 2 of 8 is 3.

In general, if $N = a^x$, then $\log_a N = x$.

The logarithm to a given base of a number is the power to which the base must be raised to make the number.

Example 1

Evaluate: a $\log_2 32$ b $\log_9 27$ c $\log_5 0.04$

a Let $N = \log_2 32$
then $2^N = 32 = 2^5$
hence $N = 5$ (equating indices)

b Let $x = \log_9 27$
then $9^x = 27$
i.e. $(3^2)^x = 3^3$
 $3^{2x} = 3^3$
hence $2x = 3$ (equating indices)
and $x = 1\frac{1}{2}$

c Let $r = \log_5 0.04$
then $5^r = 0.04 = \frac{1}{25} = 5^{-2}$
hence $r = -2$ (equating indices)

Exercise 2a

Evaluate the following logarithms.

- | | |
|-------------------------|------------------------|
| ① $\log_2 4$ | ② $\log_{10} 1\,000$ |
| ③ $\log_5 25$ | ④ $\log_3 81$ |
| ⑤ $\log_{12} 144$ | ⑥ $\log_6 216$ |
| ⑦ $\log_7 \frac{1}{7}$ | ⑧ $\log_4 64$ |
| ⑨ $\log_4 8$ | ⑩ $\log_8 4$ |
| ⑪ $\log_{25} 0.2$ | ⑫ $\log_{49} 7$ |
| ⑬ $\log_{1000} 10$ | ⑭ $\log_4 \frac{1}{8}$ |
| ⑮ $\log_{16} 0.25$ | ⑯ $\log_{100} 0.001$ |
| ⑰ $\log_9 \frac{1}{27}$ | ⑱ $\log_8 0.062\,5$ |
| ⑲ $\log_{1.2} 1.728$ | ⑳ $\log_{0.2} 25$ |

2.2 Laws of logarithms

The three fundamental laws of indices can be stated in their equivalent logarithmic form:

$$x^a \times x^b = x^{a+b}$$

$$\log(MN) = \log M + \log N$$

$$x^a \div x^b = x^{a-b}$$

$$\log \left(\frac{M}{N} \right) = \log M - \log N$$

$$(x^a)^b = x^{ab}$$

$$\log(M^p) = p \log M$$

Example 2

Prove that $\log(MN) = \log M + \log N$.

Let $\log_x M = a$ and $\log_x N = b$, where x is any base.

Then $M = x^a$ and $N = x^b$

$$\begin{aligned} MN &= x^a \times x^b \\ &= x^{a+b} \end{aligned}$$

$$\log_x(MN) = a + b = \log_x M + \log_x N$$

Notice in Example 2 that since x can take any value, the law

$$\log(MN) = \log M + \log N$$

is true for *any* base. This is true for all of the laws of logarithms.

Example 3

Given that $\log 2 = 0.301\,03$ and $\log 3 = 0.477\,12$, calculate, without using any tables

$$a \log 5 \quad b \log 6 \quad c \log 0.9$$

[WAEC]

$$\begin{aligned} a \log 5 &= \log \frac{10}{2} \\ &= \log 10 - \log 2 \\ &= 1 - 0.301\,03 \\ &= 0.698\,97 \end{aligned}$$

$$\begin{aligned} b \log 6 &= \log(2 \times 3) \\ &= \log 2 + \log 3 \\ &= 0.301\,03 + 0.477\,12 \\ &= 0.778\,15 \end{aligned}$$

$$\begin{aligned} c \log 0.9 &= \log \frac{9}{10} \\ &= \log 9 - \log 10 \\ &= \log 3^2 - 1 \\ &= 2 \log 3 - 1 \\ &= 2 \times 0.477\,12 - 1 \\ &= 0.954\,24 - 1 \\ &= -0.045\,76 \end{aligned}$$

Example 4

Simplify the following as far as possible.

$$a \log 8 + \log 5$$

$$b \quad 2 - \log 4$$

$$c \log 9 \div \log 3$$

$$d \quad \frac{2}{5} \log 32$$

$$\begin{aligned} a \log 8 + \log 5 &= \log(8 \times 5) \\ &= \log 40 \end{aligned}$$

$$\begin{aligned} b \quad 2 - \log 4 &= \log 100 - \log 4 \\ &= \log \frac{100}{4} \\ &= \log 25 \end{aligned}$$

$$\begin{aligned} c \log 9 \div \log 3 &= \frac{\log 9}{\log 3} = \frac{\log 3^2}{\log 3} = \frac{2 \log 3}{\log 3} \\ &= 2 \end{aligned}$$

$$\begin{aligned} d \quad \frac{2}{5} \log 32 &= \log 32^{\frac{2}{5}} = \log (2^5)^{\frac{2}{5}} = 2 \log 2 \\ [or] &= \log (\sqrt[5]{32})^2 = \log 2^2 \\ &= \log 4 \end{aligned}$$

Example 5

Evaluate $\log_3 6.84$ to 2 decimal places.

$$\text{Let } \log_3 6.84 = x$$

$$\text{then } 3^x = 6.84$$

$$\Leftrightarrow \log_{10}(3^x) = \log_{10} 6.84$$

$$\Leftrightarrow x \log_{10} 3 = \log_{10} 6.84$$

$$\Leftrightarrow x = \frac{\log_{10} 6.84}{\log_{10} 3}$$

working:

	No	Log
	0.8351	1.9218
	0.4771	-1.6786
	1.751	0.2432

$$= 1.75 \text{ to 2 d.p.}$$

Exercise 2b

Unless otherwise stated, assume that all the logarithms are to base 10.

- ① Given that $\log 2 = 0.301\,0$, $\log 3 = 0.477\,1$ and $\log 7 = 0.845\,1$, evaluate the following.

$$a \log 5$$

$$b \log 8$$

$$c \log 49$$

$$d \log 14$$

$$e \log 35$$

$$f \log 42$$

- ② Given that $\log 2 = 0.301\,01$, evaluate $\log 16$ without using tables.

- ③ Express the following as logarithms of single numbers.

$$a \log 3 + \log 4$$

$$b \log 15 - \log 3$$

$$c \quad 3 \log 5$$

$$d \quad \frac{1}{2} \log 25$$

- e $\log 81 - \log 3$ f $1 + \log 3$
 g $1 - \log 5$ h $2\log 3 + \log 6$
 i $2 - 2\log 5$ j $\frac{3}{4}\log 16$

④ Evaluate, without any tables,
 $3\log 2 + \log 20 - \log 1.6$. [WAEC]

⑤ Solve the following for x .

- a $\log_{10} x = 3$ b $\log_{10} x = -2$
 c $\log_x 81 = 4$ d $2\log_x (3\frac{3}{8}) = 6$

⑥ Simplify the following without using tables.

- a $\frac{\log 4}{\log 2}$ b $\frac{\log 16}{\log 8}$ c $\frac{\log 243}{\log 3}$
 d $\frac{\log \sqrt{5}}{\log 5}$ e $\frac{\log \sqrt{2}}{\log 9}$ f $\frac{\log 0.2}{\log 25}$

⑦ Simplify the following.

- a $\log 8 - \log 4$ b $\log 8 \div \log 4$
 c $\log 8 + \log 4$ d $\frac{\log 8 - \log 4}{\log 4 - \log 2}$

⑧ Express the following in index form.

- a $\log x = b$ b $\log x + \log y = 1$
 c $\log x + 1 = 0$ d $\log x + 2\log y = 3$
 e $\log x - \log y = \log z$

⑨ If $\log x - \log(2x - 1) = 1$, find x . [WAEC]

⑩ Use the method of Example 2 to prove:

- a $\log\left(\frac{M}{N}\right) = \log M - \log N$
 b $\log(M^p) = p \log M$

⑪ Use logarithm tables to evaluate the following, giving answers to two decimal places.

- a $\log_2 6$ b $\log_7 6$
 c $\log_5 12.64$ d $\log_{12} 0.8621$

⑫ Solve the following equations.

- a $(2^x)^2 - 3 \times 2^x + 2 = 0$ (Hint: let $y = 2^x$)
 b $2^{2x} - 10 \times 2^x + 16 = 0$

SUMMARY

1 The Laws of indices are:

$$x^a \times x^b = x^{a+b}$$

$$x^a \div x^b = x^{a-b}$$

$$x^0 = 1 \text{ (for all } x \neq 0\text{)}$$

$$(x^a)^b = x^{ab}$$

$$x^{-a} = \frac{1}{x^a}$$

$$x^{\frac{1}{a}} = \sqrt[a]{x}$$

$$x^{\frac{a}{b}} = \sqrt[b]{x^a} \text{ or } (\sqrt[b]{x})^a$$

These laws are true for all values of a , b and $x \neq 0$.

2 Logarithm is another word for power.
 For example:

powers	logarithms
$1000 = 10^3$	$\log 1000 = 3$
$100 = 10^2$	$\log 100 = 2$
$10 = 10^1$	$\log 10 = 1$
$1 = 10^0$	$\log 1 = 0$
$0.1 = 10^{-1}$	$\log 0.1 = -1$
$0.01 = 10^{-2}$	$\log 0.01 = -2$

Generally, if $x = a^b$, then $\log_a x = b$

3 The Laws of logarithms are:

$$\log(MN) = \log M + \log N$$

$$\log\left(\frac{M}{N}\right) = \log M - \log N$$

$$\log(M)^p = p \log M$$

2.3 Calculation using logarithm tables

Pages 139–140 of Chapter 14 contain a revision of how to use logarithm tables for calculations. Turn to those pages, study the examples 3–10 and do some of the exercises.

OBJECTIVES

By the end of this chapter you should be able to:

- 1 Solve quadratic equations using factorisation, completing the square, the quadratic formula, and graphical methods
- 2 Transform word problems into linear equations that may be solved algebraically
- 3 Transform word problems into quadratic equations that may be solved algebraically
- 4 Use algebraic and/or graphical methods to solve simultaneous linear and quadratic equations.

Teaching and learning materials

Teacher: Graph boards; aids for drawing curves (wire, twine)

Students: Graph books; aids for drawing curves (wire, twine)

3.1 Solution of quadratic equations

Books 1 and 2 of *New General Mathematics for Senior Secondary Schools* showed the following methods for solving quadratic equations:

- 1 factorisation
- 2 completing the square
- 3 use of the formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- 4 drawing graphs.

Examples 1–4 demonstrate these methods.

The method used in Example 1 involves factorising the quadratic expression in the given equation. Each of the resulting linear expressions is equated to zero and the solutions are found.

In general, if $P \times Q = 0$, then $P = 0$ or $Q = 0$ or both P and $Q = 0$.

Example 1

Solve the equation $2x^2 + 9x = 5$.

$$2x^2 + 9x = 5$$

$$\Rightarrow 2x^2 + 9x - 5 = 0$$

$$\Rightarrow (2x - 1)(x + 5) = 0$$

$$\text{Either } 2x - 1 = 0 \text{ or } x + 5 = 0$$

$$\Rightarrow 2x = 1 \quad \text{or} \quad x = -5$$

$$\Rightarrow x = \frac{1}{2} \quad \text{or} \quad x = -5$$

Sometimes a quadratic expression cannot be factorised. In such cases, other methods can be used. Example 2 shows the method of completing the square.

Example 2

Solve the equation $2x^2 - x - 8 = 0$, giving the solutions correct to two decimal places.

First, arrange the equation so that the coefficient of x^2 is 1 and the constant term is on the RHS.

$$x^2 - \frac{1}{2}x = 4$$

Make the LHS a perfect square by adding $\left(-\frac{1}{4}\right)^2$ to both sides of the equation.

$$x^2 - \frac{1}{2}x + \left(-\frac{1}{4}\right)^2 = 4 + \left(-\frac{1}{4}\right)^2$$

$$= 4\frac{1}{16}$$

$$\Rightarrow \left(x - \frac{1}{4}\right)^2 = 4\frac{1}{16}$$

$$\Rightarrow x - \frac{1}{4} = \pm \sqrt{\frac{65}{16}}$$

$$= \pm \frac{8.062}{4}$$

$$\Rightarrow x = \frac{1}{4} + \frac{8.062}{4} \text{ or } \frac{1}{4} - \frac{8.062}{4}$$

$$\Rightarrow x \approx 2.27 \text{ or } -1.77$$

Example 3 shows how to use the general formula to solve a quadratic equation. *Note:* If a quadratic equation can be factorised, do not use the formula.

Example 3

Solve the quadratic equation $5x^2 - 6x - 3 = 0$.
Give the roots correct to 2 decimal places.

Comparing

$5x^2 - 6x - 3 = 0$ with $ax^2 + bx + c = 0$,
 $a = 5, b = -6, c = -3$.

Using the formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$:

$$\begin{aligned}x &= \frac{-(-6) \pm \sqrt{(-6)^2 - 4(5)(-3)}}{2 \times 5} \\&= \frac{6 \pm \sqrt{36 + 60}}{10} \\&= \frac{6 \pm \sqrt{96}}{10} \\&\approx \frac{6 \pm 9.798}{10}\end{aligned}$$

$$\Rightarrow x \approx 1.58 \text{ or } -0.38$$

Graphical methods may also be used to solve quadratic equations, as shown in Example 4.

Example 4

Table 3.1 is an incomplete table for the relation $y = px^2 + 2x + 4$, where p is a constant.

x	-2	-1	0	1	2	3	4
y	-4	1	4				

Table 3.1

- a** Use the table to find the value of the constant p .
b Copy and complete the table.
c Using a scale of 2 cm to represent 1 unit on both axes, draw the graph of the relation $y = px^2 + 2x + 4$ for $-2 \leq x \leq 4$.
d Use your graph to obtain:
i the roots of the equation $px^2 + 2x + 4 = 0$
ii the equation of the line of symmetry of the curve. [WAEC]

- a** From the first column of the table, when $x = -2, y = -4$. Substitute these values into $y = px^2 + 2x + 4$.
 $-4 = p(-2)^2 + 2(-2) + 4$
 $\Rightarrow -4 = 4p - 4 + 4$
 $\Rightarrow -4 = 4p$
 $\Rightarrow p = -1$

- b** Since $p = -1$, the relation is $y = -x^2 + 2x + 4$.
Table 3.2 is the completed table.

x	-2	-1	0	1	2	3	4
y	-4	1	4	5	4	1	-4

Table 3.2 $y = -x^2 + 2x + 4$

- c** Fig. 3.1 shows the graph of the relation $y = px^2 + 2x + 4$ (where $p = -1$).
Note: To fit this book, the graph has been drawn to a smaller scale than required.

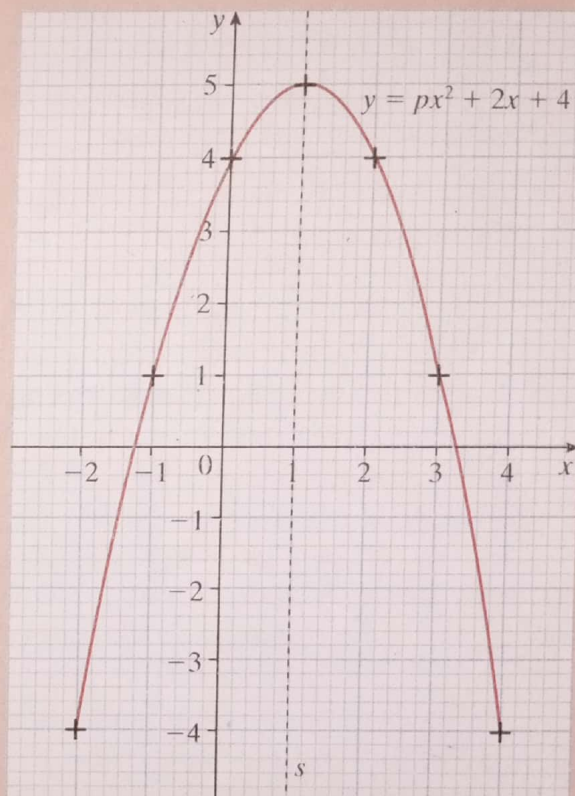


Fig. 3.1

- d** *i* The roots of the equation $px^2 + 2x + 4 = 0$ are found where the curve in Fig. 3.1 cuts the x -axis, i.e. at $x = -1.2$ and $x = 3.2$.
The roots are -1.2 and 3.2 .
ii The line of symmetry is the broken line s in Fig. 3.1.
The equation of s is $x = 1$.

In general, as Example 4 shows, the graph of $y = ax^2 + bx + c$ gives the solution of $ax^2 + bx + c = 0$ at the points where the curve cuts the x -axis.

Chapters 12 and 13 contain further examples of the graphical solution of equations.

Exercise 3a

- ① Solve the following equations by factorisation.

a $2x^2 - 9x = 0$
b $p^2 - 10p + 25 = 0$
c $b^2 + 12b + 36 = 0$
d $y^2 + 8y + 16 = 0$
e $25 = d^2$
f $9x^2 - 12x + 4 = 0$
g $4a^2 - 9 = 0$
h $121 - 4c^2 = 0$
i $3t^2 + t = 2$
j $8x^2 + 15 = 22x$
k $10y^2 + 7y = 12$
l $3b^2 = 31b + 22$
m $2(x^2 + 1) - 5x = 0$
n $12y = 3(3y^2 + 1)$

- ② Solve the following equations by completing the square. Give roots correct to two decimal places where necessary.

a $x^2 - 8x + 3 = 0$
b $y^2 + 10y + 2 = 0$
c $a^2 = a + 8$
d $t^2 - 3t = 9$
e $2x^2 - 3x = 6$
f $3x^2 + 9x = 8$
g $5x^2 + 6x + 1 = 0$
h $2x^2 - 8x + 3 = 0$

- ③ Solve the following equations using the general formula. Give roots correct to 2 decimal places.

a $x^2 - 7x + 2 = 0$ **b** $2x^2 - 10x + 7 = 0$
c $3 + x - x^2 = 0$ **d** $2x^2 = 3x + 7$

- ④ Solve the following equations, leaving the roots in surd form.

a $x^2 = 2(1 - x)$ **b** $y(y - 3) = y - 1$
c $a = \frac{a + 4}{a - 1}$ **d** $\frac{t + 1}{t + 2} + \frac{t - 3}{t - 4} = 0$

- ⑤ Check your answers to parts **b** and **c** of question 3 by drawing the graphs of:

a $y = 2x^2 - 10x + 7$ for the interval $0 \leq x \leq 5$
b $y = 3 + x - x^2$ for the range $-2 \leq x \leq 3$.

⑥ **a** Copy and complete Table 3.3 for $y = 5 + 2x - x^2$ in the range $-2 \leq x \leq 4$.

x	-2	-1	0	1	1.5	2	2.5	3	3.5	4
y		3	5				3.75			-3

Tables 3.3 $y = 5 + 2x - x^2$

- b** Taking 2 cm as 1 unit on both axes, draw the graph of $y = 5 + 2x - x^2$ for the given range.
c On the same axes, draw the graph of $y = 2$.
d Using your graphs:
 i solve the equation $5 + 2x - x^2 = 0$
 ii find the values of x for which $5 + 2x - x^2 = 2$.

3.2 Word problems leading to linear equations

Example 5

Dupe bought 7 kg of rice and 8 kg of beans for ₦3 700. In the same market Yemisi bought 8 kg of rice and 7 kg of beans for ₦3 800. If each type of food is sold at the same price per kg in the market, find the price per kg of rice and the price per kg of beans.

Let the price per kg of rice be ₦ r and the price per kg of beans be ₦ b .

From the first sentence of the question,

$$7r + 8b = 3700 \quad (1)$$

From the second sentence,

$$8r + 7b = 3800 \quad (2)$$

Multiply (1) by 8 and (2) by 7 and subtract the two equations to eliminate r .

$$56r + 64b = 29600 \quad (3)$$

$$56r + 49b = 26600 \quad (4)$$

$$15b = 3000$$

$$b = 200$$

Substituting $b = 200$ in equation (1), $r = 300$.
The prices are ₦300 per kg of rice and ₦200 per kg of beans.

Exercise 3b

- ① Divide 104 into two parts so that one part is $\frac{3}{5}$ of the other.

- 2 In an examination, Adenike obtained 19 marks more than Musa. If Adenike had obtained one and a half times her own mark, she would have scored 6 marks more than twice Musa's mark. Find the marks scored by each of them.

- 3 Fig. 3.2 shows a rectangle with its dimensions in centimetres.

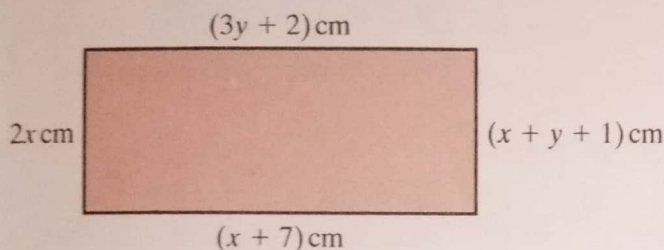


Fig. 3.2

Find:

- a the values of x and y
- b the dimensions of the rectangle.

- 4 The perimeter of a rectangular farm is 10 km. Its length is $1\frac{1}{2}$ times its breadth. Find the length and breadth of the farm.

- 5 A student cycles for x hours at 12 km/h and y hours at 16 km/h. Altogether, the student cycles 66 km in 5 hours. Find x and y .

- 6 Fig. 3.3 shows an equilateral triangle with the lengths of its sides given in terms of x and y . Find the values of x and y .

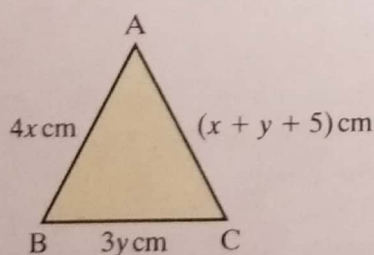


Fig. 3.3

- 7 Five years ago, a mother was twice as old as her son. In 6 years' time, the sum of their ages will be 82. Find their present ages.

- 8 The sum of the digits of a two-digit number is 11. If the digits are interchanged, the number they form decreases by 9. Find the number.

- 9 If 1 is added to the numerator and to the denominator of a fraction, the fraction becomes $\frac{3}{4}$. If 3 is added to the numerator and to the denominator the fraction becomes $\frac{4}{5}$. Find the fraction.

- 10 In a School Management Committee, men contribute ₦ x each and women ₦ y each to the maintenance fund. At one meeting, 8 men and 10 women gave a total of ₦7 000. At another meeting, 12 men and 4 women gave a total of ₦7 200. Find the values of x and y .

3.3 Word problems leading to quadratic equations

Just as some word problems lead to linear equations, so other word problems lead to quadratic equations.

Example 6

A two-digit number is such that its tens digit is greater than its units digit by 4. The two-digit number is less than 4 times the product of its digits by 11. Find the number.

Let the two-digit number be xy :

$$x = y + 4 \quad (\text{first sentence})$$

By substituting $(y + 4)$ for x we can write the number as $(y + 4)y$.

The product of its digits is $(y + 4) \times y$.

The value of the number itself is $10(y + 4) + y$

$$\Rightarrow 10(y + 4) + y = 4y(y + 4) - 11$$

(second sentence)

$$\Rightarrow 4y^2 + 5y - 51 = 0$$

$$\Rightarrow (4y + 17)(y - 3) = 0$$

$$\Rightarrow y = -\frac{17}{4} \text{ or } 3$$

$y = -\frac{17}{4}$ is unacceptable, since y must be a whole number.

$$\therefore y = 3$$

$$\text{and } x = 3 + 4 = 7$$

The number is 73.

Check: 1st sentence: $7 - 3 = 4$ ✓

$$\text{2nd sentence: } 73 = 4(7 \times 3) - 11$$

$$= 84 - 11$$

$$= 73 \quad \checkmark$$

Example 7

A lorry travels a distance of 240 km at a uniform speed. If it had been 4 km/h slower, it would have taken 2 hours more to cover the distance. Find its speed.

Let the speed be x km/h.

$$\text{Time taken} = \frac{240}{x} \text{ hours}$$

If the lorry travels 4 km/h slower, the speed is $(x - 4)$ km/h.

$$\text{Time taken} = \frac{240}{x - 4} \text{ hours}$$

$$\text{But } \frac{240}{x - 4} - \frac{240}{x} = 2 \text{ (second sentence)}$$

$$\Rightarrow 240x - 240(x - 4) = 2(x)(x - 4)$$

$$\Rightarrow 960 = 2x^2 - 8x$$

$$\Rightarrow x^2 - 4x - 480 = 0$$

$$\Rightarrow (x + 20)(x - 24) = 0$$

$$\Rightarrow x = -20 \text{ or } 24$$

$x = -20$ is unacceptable since the speed is not negative.

$$\therefore x = 24$$

The speed of the lorry is 24 km/h.

Exercise 3c

- ① A street trader spent ₦550 on x articles, each costing ₦ $(2x - 28)$. Find x .
- ② Two numbers differ by 2. The sum of their squares is 244. Find the numbers.
- ③ A rectangular farm is one hectare in area. Its perimeter is 500 metres. Find the dimensions of the farm in metres.
- ④ The length of a room is 3 metres longer than its breadth. The area of the room is 28 m^2 . Find its dimensions.
- ⑤ The product of two consecutive odd numbers is 323. Find the numbers.
- ⑥ If a number is increased by 30, it is less than its square by 12. Find the number.
- ⑦ A cyclist travels from A to B, a distance of 8 km, at x km/h. He travels from B to A, the same distance, at $x - 4$ km/h. If the total time taken is 1 hour 10 minutes, find x .

- ⑧ The dimensions of the right-angled triangle ABC in Fig. 3.4 are given in terms of x . Find x .

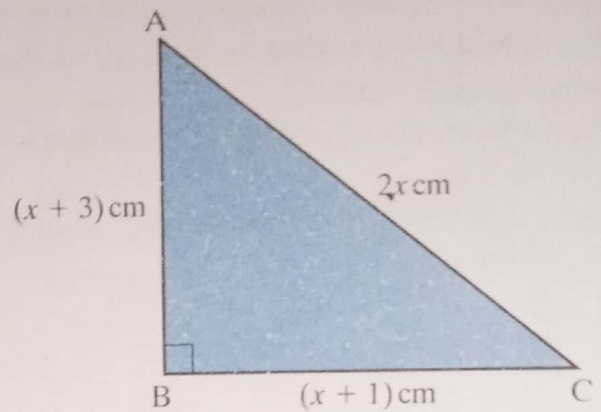


Fig. 3.4

- ⑨ The right-angled triangle in Fig. 3.5 has dimensions as shown. Find x .

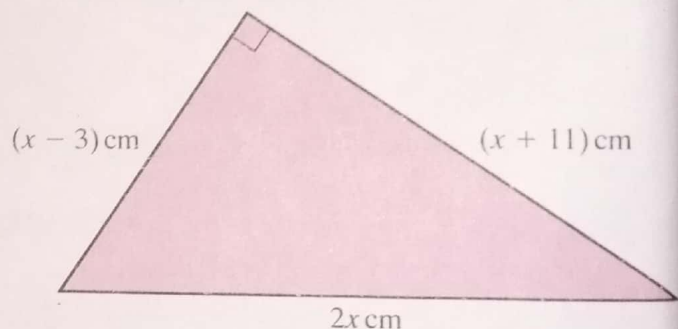


Fig. 3.5

- ⑩ Two numbers differ by 7. If their product is 120, find the numbers.
- ⑪ If 13 times a number, x , is added to twice its square, the result is 84.
a Obtain a quadratic equation in x .
b Find x .
- ⑫ If 13 times a number is subtracted from the square of the number, the result is 30. Find the number.
- ⑬ The length of a rectangle is twice its width. If the width is reduced by 1 cm and the length is also reduced by 2 cm, the area will be 15 cm^2 . Find the dimensions of the original rectangle.
- ⑭ James walks a distance of 10 km at an average speed of x km/h. Jane walks the same distance at $(x - 1)$ km/h. If the difference in their times is 30 minutes, find the value of x .